

Hierarchical floorplanning

Hierarchy packs cluster AB on the left and cluster CDE on the right with x offset five

Pseudocode

- Hierarchical floorplan packs inside clusters first
- Open this module's examples file and find the Pseudocode section
- That written sketch is what you implement on the implement track and what the browser

Algorithm sketch

- Teaching golden places cluster A B on the left and C D E on the right starting at x equals

Algorithm sketch — try these

INPUT: clusters of modules

OUTPUT: top-level pack of cluster bboxes

pack each cluster internally (slice/B*/SA)

pack cluster bboxes in outline

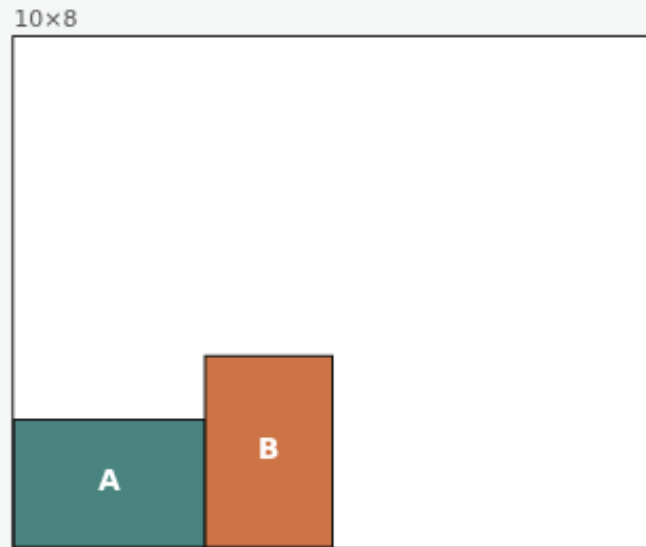
GOLDEN teaching: AB left; CDE right @x=5

Two clusters: AB and CDE

HIERARCHICAL FLOORPLAN

Two clusters: AB and CDE

Step 1 / 5 · clusters · module04-01-hierarchical-floorplan



Explanation

Hierarchy packs locally first. Left cluster holds A and B; right cluster holds C, D, and E. Each cluster is a mini-floorplan.

- Left: AB
- Right: CDE
- Pack then place clusters

clusters: 2

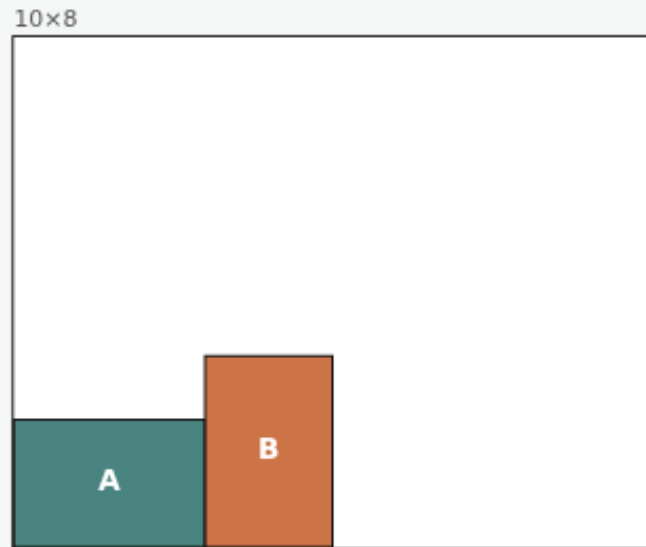
Two clusters: AB and CDE

Left cluster packs at $x < 5$

HIERARCHICAL FLOORPLAN

Left cluster packs at $x < 5$

Step 2 / 5 · left · module04-01-hierarchical-floorplan



Explanation

AB occupies the left side with A at the origin and B at x equals three. Everything in this cluster stays left of x equals five.

- $A.x < 5$
- $B.x < 5$
- Local packing done

$$\text{left max } x+w \leq 5$$

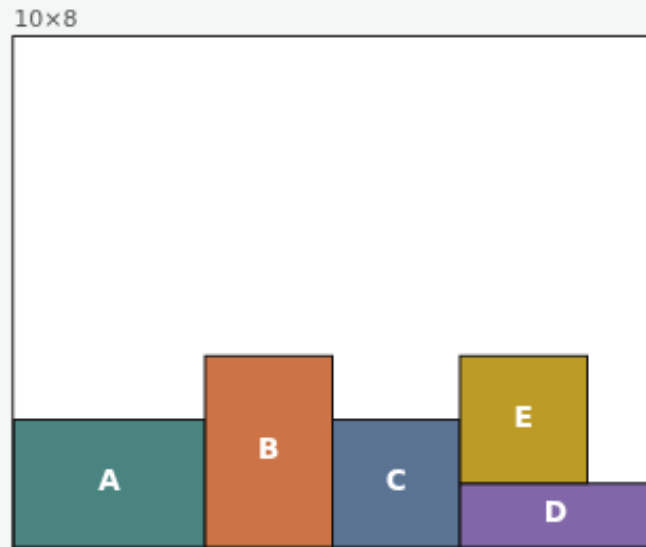
Left cluster packs at $x < 5$

Right cluster offsets by 5

HIERARCHICAL FLOORPLAN

Right cluster offsets by 5

Step 3 / 5 · right · module04-01-hierarchical-floorplan



Explanation

CDE packs in local coordinates, then shifts by five in x. C, D, and E all land at x greater than or equal to five—no cluster overlap.

- offset $x=5$
- $C.x \geq 5$
- $E.x \geq 5$

`rightMin x ≥ 5`

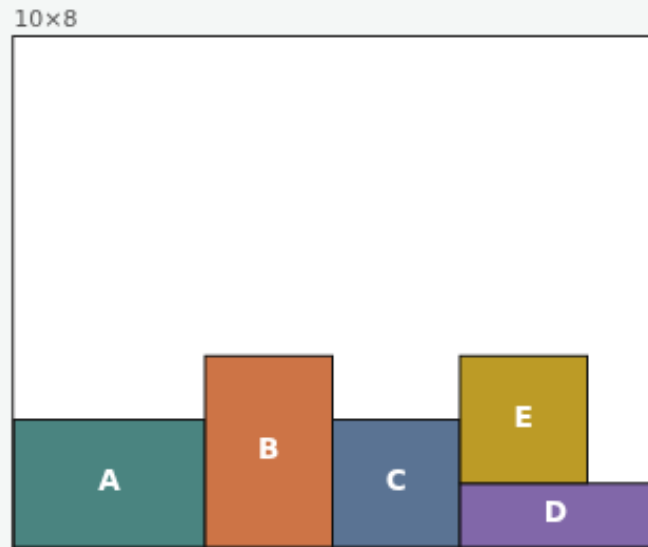
Right cluster offsets by 5

Hierarchy is legal overall

HIERARCHICAL FLOORPLAN

Hierarchy is legal overall

Step 4 / 5 · legal · module04-01-hierarchical-floorplan



Explanation

Placing the two cluster bounding boxes yields a legal chip packing. Hierarchy scales: recurse inside clusters, then assemble.

- Five modules placed
- legal: true
- Clusters separated in x

legal: true

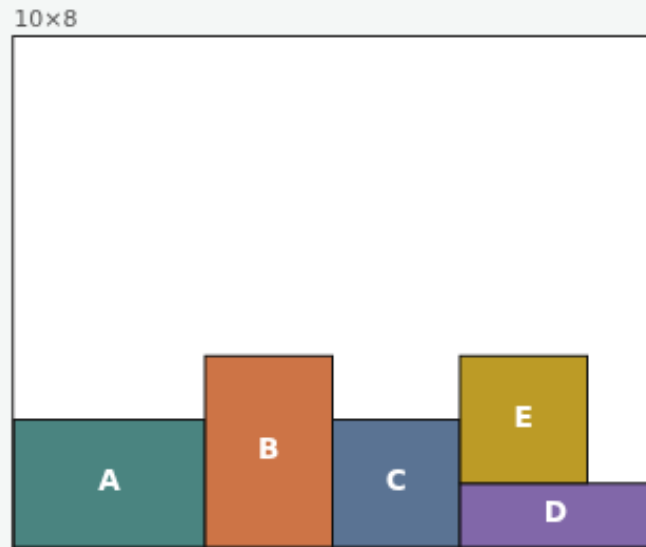
Hierarchy is legal overall

Hierarchy reuses the same engines

HIERARCHICAL FLOORPLAN

Hierarchy reuses the same engines

Step 5 / 5 · takeaway · module04-01-hierarchical-floorplan



Explanation

Each level can use slicing, B-star, or SA. The teaching golden is simply AB left and CDE right at offset five.

- Pack locally, place globally
- Same legality rules
- Next: pin assignment

starter: AB | CDE @x=5

Hierarchy reuses the same engines

Browser lab track

- Open hierarchical-floorplan and Pack hierarchy
- Confirm A and B have x less than five, C D E have x at least five, and legality true

Implement track

- Implement `pack_hierarchical`
- Assert `leftMax x+w` is at most `rightMin x`, five modules present, and `legality true`

Pitfalls

- Overlapping cluster bounding boxes

Your turn

- Accept the AB | CDE packing
- Next: assign pins P0 through P3 on all four outline sides